Texture Lab: Webcam Grid

CS 5382 Computer Graphics • Spring 2026 • Daniel Oni

What the program does

The page opens my webcam, uploads each frame to a WebGL texture, and draws the feed as a 4 x 4 grid of tiles on the canvas. Starting point was the `CS5382_WebcamStarter.html` file from Canvas, and the grid effect is produced by only two small changes to that starter.

How I achieved the grid effect

I did not draw sixteen separate quads. The starter already draws one quad that covers the whole canvas and samples the webcam texture on it. I only changed two things about how that quad samples the texture:

1. The texture coordinates on the quad run from 0 to 4 on both the S and T axes instead of the usual 0 to 1. So the corners of the quad get (0, 0), (4, 0), (4, 4), and (0, 4).

2. The wrap mode for the texture is set to `gl.REPEAT` on both `TEXTURE_WRAP_S` and `TEXTURE_WRAP_T` (the starter had these as `CLAMP_TO_EDGE`). With REPEAT, any coordinate outside the 0 to 1 range simply wraps back around, so the image tiles four times on each axis automatically and I get a 4 x 4 grid for free.

I also turned on `UNPACK_FLIP_Y_WEBGL` before uploading video frames so the image appears upright, and I pass null to the initial `texImage2D` call because at that moment the camera stream has not yet attached to the video element. The grid size is controlled by a single variable `N` near the top of the file, so it takes one edit to switch between a 4 x 4 and a 3 x 3 grid.

Files submitted

- `CG_TextureLab_WebcamGrid_OniDaniel.html` (the modified starter)
- A screenshot of the 4 x 4 grid running on my webcam
- This writeup

Note on running the file

The HTML references `initShaders.js` the same way the starter does, so both files need to sit in the same folder when it is served. Browsers only give camera access to pages served over a secure origin, so I ran the HTML through a local server (`python -m http.server` in the folder, then opened `http://localhost:8000`) rather than opening the file directly with `file://`.